

Lecture 1: Fixed- and Floating-point Numbers

Their Properties and Exponent Bias

CSE 330 — Numerical Methods

1 Fixed Point

A fixed-point number has a fixed position for the radix point, separating the *whole* part from the *fraction* part:

$$x = \pm \left(\underbrace{d_1 d_2 d_3 \dots d_{n-1}}_{\text{whole}} . \underbrace{d_n d_{n+1} \dots d_k}_{\text{fraction}} \right)_\beta$$

Example.

$$\begin{aligned}(10.1)_2 &= 1.01 \times 2^1 = 0.101 \times 2^2 \\ (10.25)_{10} &= 1.025 \times 10^1 = 0.1025 \times 10^2\end{aligned}$$

2 Floating Point

In floating point the radix point “floats” — the value is written with a fraction times a power of the base:

$$0.1025 \times 10^2 \equiv 10.25$$

3 Defining a Floating Point Number

A floating-point system is a subset $F \subset \mathbb{R}$ whose elements are of the form

$$F = \pm (0.d_1 d_2 \dots d_m)_\beta \times \beta^e$$

- $\pm$: sign
- $(0.d_1 d_2 \dots d_m)$: fraction / significand / **mantissa**
- β : base
- e : exponent, where $e_{\min} \leq e \leq e_{\max}$ (always)
- m : number of fraction digits after the point “.”

4 Worked Example: $\beta = 2$, $m = 3$, $e \in [-1, 2]$

The representable numbers look like

$$\pm \left(0. \square \square \square \right)_2 \times 2^\square, \quad \text{with } e_{\max} = 2.$$

Largest (max):

$$(0.111)_2 \times 2^2 = \frac{7}{2}$$

Smallest non-zero (min):

$$(0.100)_2 \times 2^{-1} = \frac{1}{4}$$

A value larger than the maximum causes **overflow**; a value smaller than the minimum causes **underflow**.



Counting the representable numbers

$$\underbrace{2}_{\text{sign } \pm} \times \underbrace{2^{m-1}}_{\substack{\text{free mantissa bits} \\ \text{(always starts with .1)}}} \times \underbrace{|e|}_{\text{exponent count}} = 2^m \times |e|$$

$$= 2^3 \times |4| = 8 \times 4 = 32 \text{ values.}$$

5 Three Types of Convention

In every convention the leading significant digit is always 1 (mantissa = d).

- i. **Standard form:** $\pm (0.d_1 d_2 \dots d_m)_\beta \times \beta^e$
- ii. **Denormalized form:** $\pm (1.d_1 d_2 \dots d_m)_\beta \times \beta^e$
- iii. **Normalized form:** $\pm (0.1 d_1 d_2 \dots d_m)_\beta \times \beta^e$

Take $\beta = 2$, $m = 3$, $e \in [-1, 2] = \{-1, 0, 1, 2\}$.

(i) Standard form.

$$\pm (0.1 _ _)_2 \times 2^e$$

Free mantissa bits give $2^2 = 4$ patterns, and $|e| = 4$, so

$$2^2 \times 4 = 16 \text{ total (per sign).}$$

(ii) Normalized form. The first digit (the leading 1) is implicit and *not* stored as mantissa.

Extreme values.

$$\text{Smallest non-negative: } (0.1 _ _)_2 \times 2^{-1} = \frac{1}{4}$$

$$\text{Highest: } (0.1111)_2 \times 2^2 = \frac{15}{4}$$

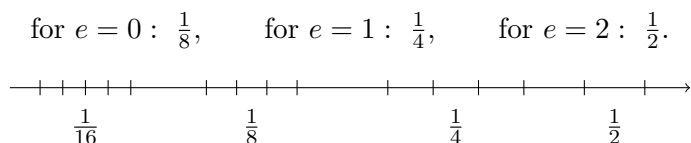
$$\text{Smallest (most negative) value: } -\frac{15}{4}$$

6 All Values Under Convention (i): $\beta = 2$, $m = 3$, $e \in [-1, 2]$ Standard form

For $e = -1$:

$$\begin{aligned} (0.100)_2 \times 2^{-1} &= \frac{1}{4} \\ (0.101)_2 \times 2^{-1} &= \frac{5}{16} \\ (0.110)_2 \times 2^{-1} &= \frac{3}{8} \\ (0.111)_2 \times 2^{-1} &= \frac{7}{16} \end{aligned} \quad (\text{gap} = \frac{1}{16} \text{ each})$$

The spacing between consecutive numbers **grows with the exponent**:



7 IEEE Standard (1985) — Double Precision

A 64-bit number representation, laid out as:

sign	exponent	mantissa
1	11 (range)	52 (precision)

- 1 bit — sign
- 11 bits — exponent / range, giving $e \in (0 - 2047)$
- 52 bits — precision / mantissa

Denormalize:

$$(1.d_1d_2 \dots d_{m=52})_\beta \times 2^e$$

Largest:

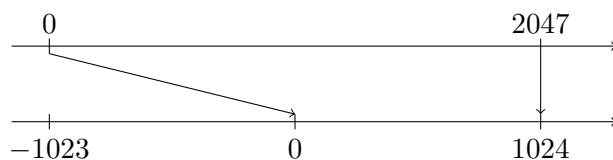
$$(1.11111 \dots 1)_2 \times 2^{2047} \Rightarrow \text{too large, not practical}$$

Smallest:

$$(1.000 \dots 0)_2 \times 2^0 \Rightarrow \text{negative exponent values??}$$

8 Exponent Biasing

Idea: scale the exponent *down* so it can take negative values. The stored range $e \in [0, 2047]$ is shifted by a **bias of 1023**:



With biasing the conventions become:

$$\mathbf{Denormalized:} \quad (1.d_1d_2\dots d_m)_2 \times 2^{e-1023}$$

$$\mathbf{Normalized:} \quad (0.1\,d_1d_2\dots d_m)_2 \times 2^{e-1022} \quad (\text{needed})$$

So the effective exponent range becomes [Calculatively]

$$e \in [-1022, 1025]$$